Proof of the 3D version of the Euler Inequality

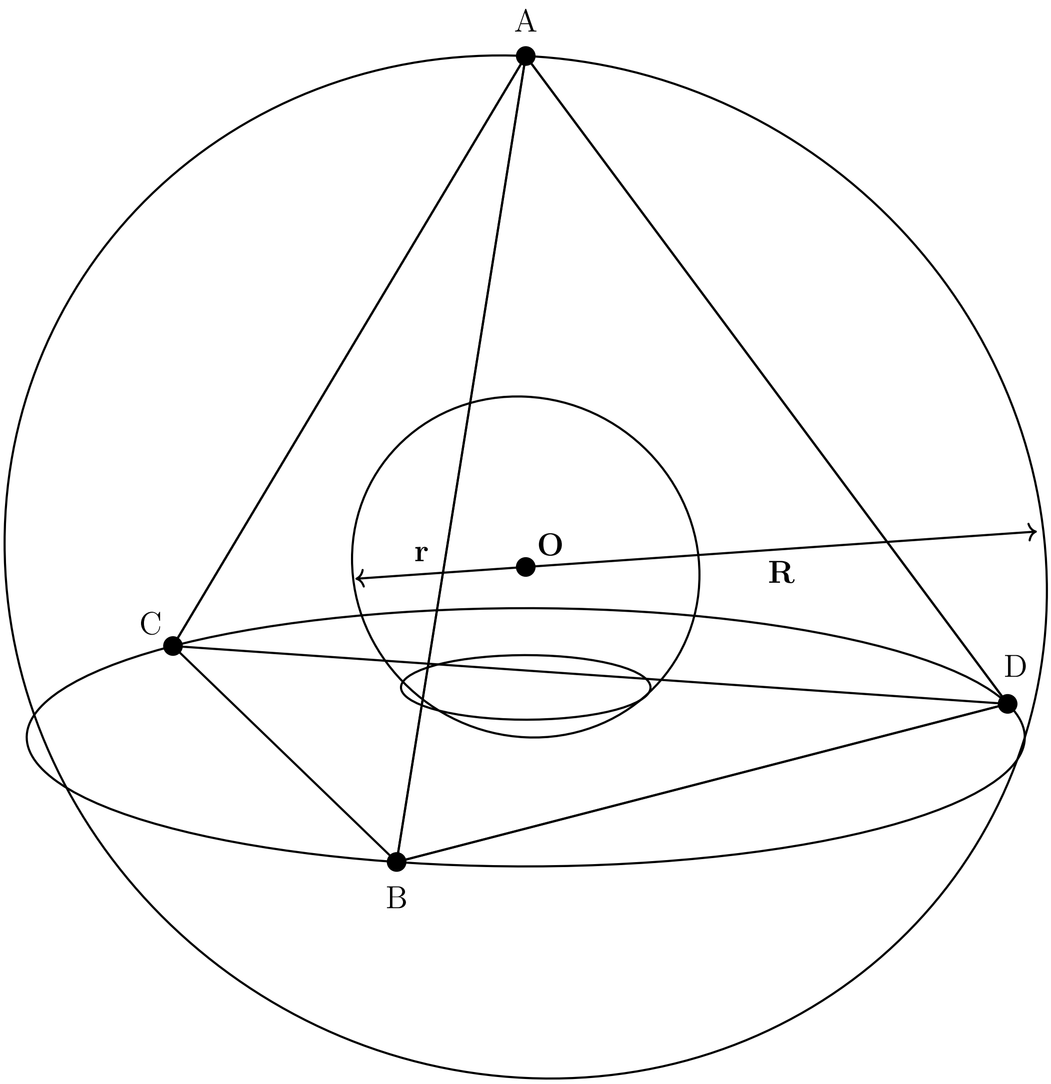

Diagram of the inscribed and circumscribed sphere of a tetrahedron

Contents

1 Formal Statements of the Result

Theorem 1.1 (3D Euler Inequality)**.** *Let points A, B, C, D denote the vertices of a tetrahedron in $\mathbb{R}^3$; let R, r be the radii of the circumscribed and inscribed sphere respectively. Then,*

$$\forall A, B, C, D \in \mathbb{R}^3, R \geq 3r.$$

2 Proof

Proof of the inequality. We will first define the vector $\overrightarrow{OI}$, where I is the center of the inscribed sphere, in terms of the vectors $\underset{\sim}{a}$, $\underset{\sim}{b}$, $\underset{\sim}{c}$, $\underset{\sim}{d}$ as the vectors to A, B, C, D from the origin respectively. $\square$